

Date: 3/09/25 Task - 6 (a)

Implement various text file operations

(a) Student Record file Handling

AIM:

To write a Python program that creates a text file to store student names with their marks, read and display student records, and append new student data to the same file.

ALGORITHM:

1. Start the program.
2. create a text file (students.txt) and write student names with their marks.
3. Read all student records from the file and display them.
4. Append new student records to the same file.
5. Read again to show updated records.
6. End the program.

Program -

```
def create_file():  
    with open("students.txt", "w") as  
f:  
    f.write("Alice 85\n")  
    f.write("Bob 90\n")  
    f.write("Charlie 78\n")  
print("File created and initial records stored.")
```

```
def read_file():  
    print("\nStudent Records:")  
    with open("students.txt", "r") as  
f:  
    for line in f:  
        print(line.rstrip())
```


Sample Output

File created and initial records stored.

Student Records:

Alice 85

Bob 90

Charlie 78

New student data appended.

Student Records:

Alice 85

Bob 90

Charlie 78

David 88

Eva 92.



```
def append - filec):
```

```
with open("Students.txt", "a") as
```

```
f:
```

```
    f.write("David 88\n")
```

```
    f.write("Eva 92\n")
```

```
    print("\n New student data appended")
```

RESULT

Implementation of the Students records file and performing the text file operations are executed successfully.

3/09/21 Task - 6 (b)

TEXT FILE WORD ANALYZER

AIM:

TO write a python program that reads a text file, counts the number of lines, words, and characters, and displays the frequency of each word.

ALGORITHM

1. Start the program
2. Open a text file (Sample.txt) in read mode.
3. Count number of lines, words, and characters
4. Use a dictionary to store frequency of each word.
5. Display counts and word frequency.
6. End the program.

Program

```
def analyze_file(file_name):
```

```
    with open(file_name, "r") as f:
```

```
        text = f.read()
```

```
    # count lines, words, characters
```

```
    lines = text.splitlines()
```

```
    word = text.split()
```

```
    characters = len(text)
```

```
    print(f"Number of lines: {len(lines)}")
```

```
    print(f"Number of words: {len(word)}")
```

```
    print(f"Number of characters: {characters}")
```

Sample Output

Number of lines : 2

Number of words : 5

Number of characters : 33

Word Frequency:

hello : 2

World : 2

pythoh : 1



word frequency

freq = {}

for word in words:

word =

word.lower().strip(".,!?"')

freq[word] = freq.get(word, 0) + 1

print("\nWord Frequency:")

for word, count in freq.items():

print(f"{word} : {count}")

Execution

First create a file

with open("sample.txt", "w") as f:

f.write("HelloWorld\n")

f.write("Hello Python World\n")

VEL TECH - CSE	
EX NO.	6
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	25
WITH DATE	

RESULT: python program to read and perform count of no. of words and characters & display word frequency in textfiles is implemented & executed successfully.

17/4/25